Deep Learning-Based Document Image Quality Enhancement

Research Project Proposal • Bijon Mollick • Computer Science & Engineering

Proposed by	Bijon Mollick
Project type	Research / prototype study
Estimated duration	8–10 weeks
Max collaborators	3

Abstract

I want to investigate how deep learning techniques can improve the quality of scanned and photographed documents before they are processed by downstream systems. The project will focus on noise removal, contrast enhancement, blur reduction, and image restoration, with the goal of making low-quality documents easier for computer vision and OCR applications to process.

Research Motivation

Document images captured by mobile cameras or older scanners often contain shadows, blur, uneven illumination, compression artifacts, and background noise. These problems can reduce the quality of later document-analysis tasks. I want to study whether a lightweight enhancement pipeline can improve visual quality in a measurable and reproducible way.

Research direction: I will prioritize practical experimentation and clear evaluation. Rather than developing an unnecessarily large model, I want to compare a manageable baseline with a deep-learning-based enhancement approach and identify where the improvement is actually useful.

Research Objectives

- Study common quality problems in scanned and photographed document images.
- Prepare a suitable dataset containing clean and degraded document images.
- Develop a deep-learning-based document enhancement model or pipeline.
- Evaluate whether enhancement improves image quality and downstream OCR performance.
- Analyze failure cases such as severe blur, shadows, and very low contrast.

Research Questions

RQ1. Which types of document degradation have the largest effect on image readability?

RQ2. How much improvement can a lightweight deep-learning enhancement model provide?

RQ3. Does visual enhancement consistently improve OCR performance?

RQ4. Which degradation types remain difficult even after enhancement?

Proposed Methodology

Step 1. Review recent approaches for document image enhancement, restoration, and OCR preprocessing.

Step 2. Select or construct an appropriately licensed dataset of document images and controlled degradations.

Step 3. Apply preprocessing such as resizing, normalization, and controlled noise/blur generation.

Step 4. Establish a simple image-processing baseline before introducing the learning-based method.

Step 5. Train or fine-tune a suitable lightweight deep-learning model for enhancement.

Step 6. Compare enhanced images with degraded inputs using image-quality metrics and visual inspection.

Step 7. Run an OCR experiment where appropriate to determine whether enhancement provides practical downstream benefits.

Step 8. Perform error analysis and document cases where the model introduces artifacts or fails to recover useful information.

Data and Experimental Resources

The study will use public/licensed document images or a small consent-based dataset. Degraded versions can be generated in a controlled way so that the type and severity of corruption are known. The main experiments can be performed in Python using a notebook-based workflow, with GPU access used only if required by the selected model.

Evaluation Plan

Evaluation will combine objective image-quality measurements with qualitative inspection. Depending on the available reference images, metrics such as PSNR and SSIM may be used. A small OCR comparison can also be included to test whether visual improvements translate into better text extraction. Results will be reported across different degradation types rather than using only one overall score.

Expected Challenges

- Real document degradation can be more complex than synthetic corruption.
- A model may improve visual appearance while unintentionally removing useful text details.
- A small dataset may limit the generalization of the trained model.
- OCR improvement may depend on the OCR engine as well as the enhancement model.

Reproducibility and Documentation

I will document dataset sources, degradation settings, preprocessing decisions, model configuration, training settings, evaluation metrics, and important observations. The aim is to keep the core experiment reproducible so that a collaborator can review the methodology and repeat the main comparison.

Planned Mitigation

I will begin with a small baseline and a feasibility experiment. If training a larger model becomes impractical, the project will be narrowed to a lightweight model and a carefully defined set of degradation types. Negative or inconclusive results will also be reported rather than excluded.

Expected Outcomes

- A documented dataset/preprocessing pipeline for degraded document images.
- A working deep-learning-based document enhancement prototype.
- Quantitative and qualitative comparison between baseline and enhanced images.
- An analysis of whether enhancement improves downstream OCR.
- A research report describing findings, limitations, and future improvements.

Interest Tags

Machine Learning • Artificial Intelligence • Image Processing • Document Analysis • Deep Learning • Image Restoration

Required Skills

Python • TensorFlow • Deep Learning • OpenCV • Image Processing

Technical Stack

Python, TensorFlow or PyTorch, OpenCV, NumPy, Jupyter Notebook, and Git.

Potential Research Collaboration

I am looking for collaborators who are interested in deep learning, image processing, or computer vision. A collaborator could contribute to dataset preparation, model selection, experiment design, OCR evaluation, or analysis of the results.

Scope and Responsible Research

Only public, appropriately licensed, or consent-based document images will be used. Personal information contained in documents will be avoided or removed where possible. The project is intended as an academic research prototype and will not claim production-level document restoration performance without sufficient evidence.

Proposed Work Plan

Phase	Activities
1. Background study	Literature review, problem definition, dataset selection
2. Data preparation	Clean images, define degradation types, create experimental splits
3. Baseline	Implement and evaluate a simple enhancement baseline
4. Model & experiments	Train/fine-tune model and compare approaches
5. Evaluation	Image metrics, OCR comparison, error analysis
6. Reporting	Results, limitations, documentation, presentation

What I Am Looking for in a Collaborator

I would like to work with someone who is genuinely interested in the research question and can contribute to experimentation, technical discussion, dataset preparation, or academic writing. I am open to refining the scope after discussing feasibility and available resources.

Possible Next Step

The first step would be to review the research question and dataset options together, agree on a realistic experimental scope, and define responsibilities and milestones.

Note: This document describes a proposed research project. It does not claim completed research, published results, or guaranteed performance. The methodology and scope can be refined through collaboration.